

Module name: ME2171

Semester: 4

Reverse Engineering of a Soldering Iron

By

Index No.	Name	Contribution
230400H	V. L. R. Matharaarachchi	20%
230005C	S. W Abeykoon	20%
230008M	A. L. Abeysinghe	20%
230120A	E. D. T. N De Alwis	20%
230393K	U. I. Mallawaarachchi	20%

Date of submission	14/05/2026
Due date of submission	14/05/2026

**Department of Mechanical Engineering
University of Moratuwa
Sri Lanka**

Table of Contents

Table of Contents	1
1 Introduction	4
2 Components	5
3 Part No: 1 Soldering Iron Tip - 230400H	7
3.1 Component Overview	7
3.2 Task A - Product Analysis	7
3.2.1 Identification of the material	7
3.2.2 Final Conclusion, Material Identification and Justification	10
3.2.3 Manufacturing Process	11
3.3 Task B – Discussing the feasibility of Casting	11
3.4 Task C – Discussing the feasibility of Forming	12
4 Part No 2: Metal Barrel - 230005C	13
4.1 Introduction	13
4.2 Task A - Product Analysis	13
4.2.1 Identification of the Material	13
4.3 Task B - Casting feasibility	16
4.4 Task C - Forming	16
4.4.1 Possibility and Appropriateness	16
4.4.2 Likely Forming Processes and Selection	16
4.5 Identification of the Manufacturing Process	18
4.5.1 Potential Manufacturing Processes and Justification	18

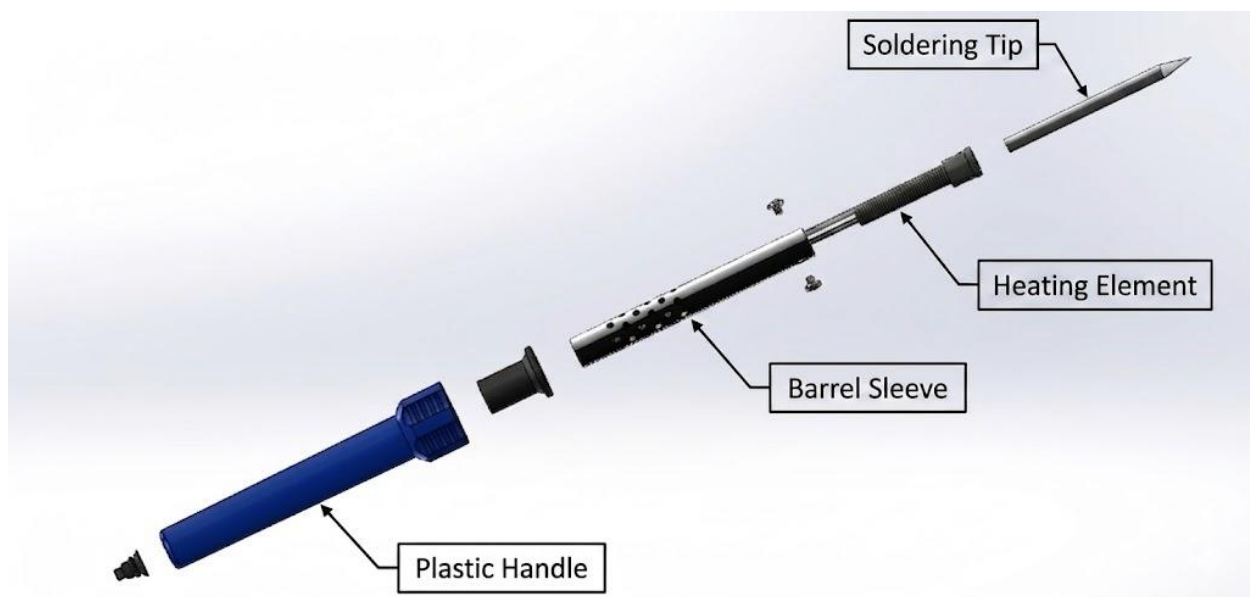
4.5.2 Selected Process – Precision Stamping followed by Roll Forming	18
5 Part No 3: Ceramic Part - 230008M	19
5.1 Introduction	19
5.2 Task A - Product Analysis	19
5.2.1 Identification of the Material	19
5.2.2 Final Conclusion, Material Identification and Justification	24
5.3 Task D - Non-Metallic Manufacturing	24
5.3.1 Possibility and Appropriateness	24
Suitable Manufacturing Method – Dry Pressing	24
5.3.2 Potential Manufacturing Processes and Justification	25
Selected Process – Dry Pressing	25
Justification	26
6 Part No 3: Plastic Handle - 230393K	27
6.1 Component Overview	27
6.2 Task A - Product Analysis	27
6.2.1 Identification of the material	27
6.2.2 Final Conclusion, Material Identification and Justification	29
6.2.3 Manufacturing Process	30
6.3 Task D - Non-Metallic Manufacturing	31
6.3.1 Possibility and Appropriateness	31
6.4 Potential Manufacturing Processes and Justification	31
6.5 Proposed Modification and Improvements	32
7 Part No 5: Heating Element - 230120A	33

7.1 Identification of the Manufacturing Process	36
7.2 Task B - Casting feasibility	37
7.3 Task C - Forming feasibility	37
7.4 Non-Metallic Manufacturing	38
8 Production Drawings	39
9 References	44

1 Introduction

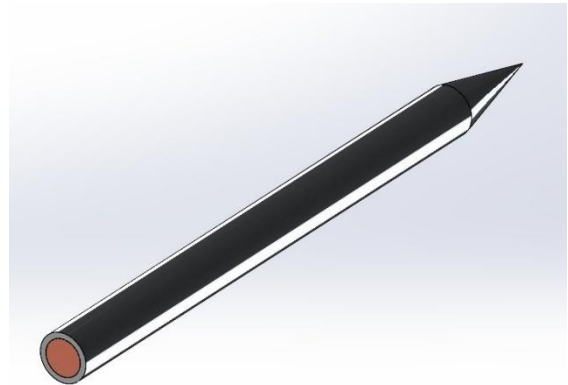
A soldering iron is a handheld tool used to join metallic components by melting a filler metal known as solder. It converts electrical energy into heat and transfers that heat through a conductive tip to create reliable electrical and mechanical joints. Soldering irons play a critical role in a wide range of industries and engineering applications, including electronics manufacturing, circuit board assembly, automotive electronics, telecommunications, aerospace systems, appliance repair, and precision prototyping. They are essential for creating durable electrical connections in components such as printed circuit boards (PCBs), sensors, wiring systems, and microelectronic devices. The quality and efficiency of soldering directly affect product reliability, electrical performance, and manufacturing productivity.

Reverse engineering a soldering iron is important because it enables engineers to analyse the design, materials, manufacturing methods, thermal behaviour, and performance characteristics of the product. Through reverse engineering, weaknesses such as poor heat transfer, rapid tip degradation, inefficient energy usage, or high production cost can be identified and improved. It also helps in understanding industry-standard design practices, optimizing component selection, improving manufacturability, and developing innovative or cost-effective alternatives

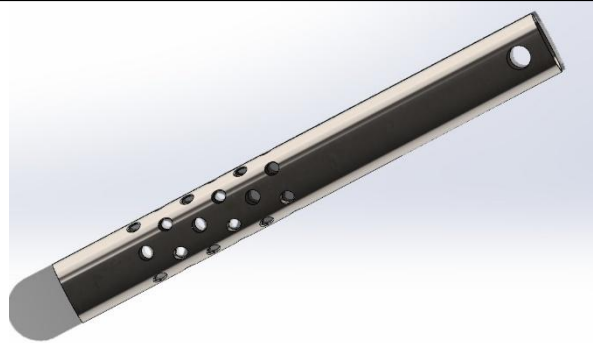


2 Components

1. Soldering Iron Tip (230400H)



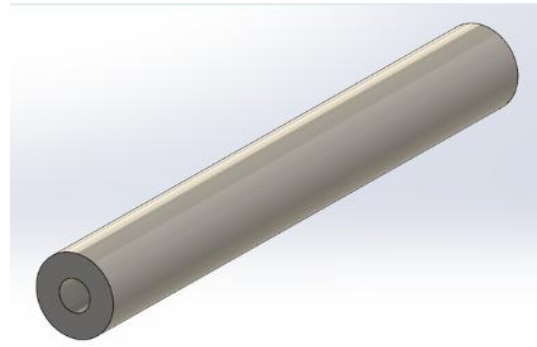
2. Metal Barrel (230005C)



3. Heating Element (230120A)



4. Ceramic Component (230008M)



5. Plastic Handle (230393K)



3 Part No: 1 Soldering Iron Tip - 230400H

3.1 Component Overview

A soldering tip is the functional end component of a soldering iron responsible for transferring heat from the heating element to the solder and the workpiece during the soldering process. It enables efficient melting and flow of solder to create reliable electrical and mechanical joints. The tip also determines the precision, responsiveness, and overall effectiveness of the soldering operation depending on its shape and thermal behaviour.



To perform effectively, a soldering tip requires several important material properties. High thermal conductivity is essential for rapid and efficient heat transfer, while good wear resistance is needed to withstand continuous contact with molten solder and components. Oxidation and corrosion resistance help maintain consistent solder wetting and prolong service life under elevated temperatures. Additionally, sufficient hardness and mechanical strength are important to prevent deformation and maintain dimensional accuracy during prolonged usage. These properties collectively ensure stable performance, durability, and reliable solder joint formation.

3.2 Task A - Product Analysis

3.2.1 Identification of the material

During the study, various standard and non-standard tests were conducted with the motive of finding out the material from which the respective component i.e., the soldering iron tip, was made. However, the intention was not to conclude on the actual material that it was made but to find out the material sub class it belonged to. Various tests that were conducted with the aforementioned purposes will be examined in this section.

(i) Visual Inspection (Non-standard Test)

Reasons for Selecting

Visual inspection is a basic and convenient material characterization method used to evaluate the surface condition and various direct qualities of a soldering tip. It helps identify characteristics such as oxidation, wear, coating damage and deformation etc at a glance.

Observations

- Surface was rough and grainy, indicating likely surface corrosion.
- Tip was conical with a worn, rounded end owing to consistent usage.
- Overall color was dull grey, suggesting oxidation of the outer surface.
- Cross-section revealed a pinkish-brown core (copper-like) with a silvery outer coating (steel-like).

Conclusion

The tip appears to be made of two metallic materials. Probably being a copper-based core enclosed within a steel outer coating. The inner core showed optical characteristics typical of copper alloys, while the coating exhibited properties comparable to steel.

(ii) Rockwell C Hardness test

Reasons for selecting

Rockwell hardness testing was selected because it provides a quick, direct measurement of surface hardness without requiring complex specimen preparation. It has the potential of directing the study into finding the material that the outer coating was made with.

Observations

Measured hardness: HRC 60 - 37.5



Conclusion

This range of hardness is only achievable with iron-based alloys, steel, or a hard nickel layer. Copper and copper alloys cannot be meaningfully tested under the Rockwell C scale they are far too soft to register in this range. This fact points to a steel or hard-metal outer layer.

(iii) Density test

Reasons for selecting

A density test using a density bottle is suitable for a soldering iron tip because it allows accurate measurement of density through the displacement method without requiring complex equipment. And the values thus obtained could be used in comparing with the documented density values of the material.

Observations

	Mass (g)
Bottle	16.7703
Bottle + Specimens	17.2814
Bottle + Water + Specimen	74.0191
Bottle + Water	73.5677
Density (kg/m ³)	8561.14



Thereby the observed density was 8561.14 kgm⁻³

Conclusion

The measured density of 8,561 kg/m³ lies closer to but not exact to the density of pure copper which is 8960 kg/m³. This value is consistent not with pure copper but with a copper containing significant copper content, supporting the visual inspection findings.

(iv) Spark Test

Reasons for Selecting

The spark test was conducted with the expectation of getting an insight of the metallic material of the soldering iron tip by observing the nature of the spark.

Observation

- No well-defined sparks were produced
- Only a few short, inconsistent sparks were observed

Conclusion

The absence of proper sparks provides further support for a copper-dominant material. Copper and copper alloys produce little to no spark reaction during grinding, unlike ferrous metals which generate bright, branching sparks. This corroborates the density and visual findings.



3.2.2 Final Conclusion, Material Identification and Justification

By considering the high thermal conductivity the tip should display in its functionality, and by the fact that the experimental density of the core material is 8561 kg/m³ the core should be a metal or alloy containing a greater amount of copper. However, it cannot be pure copper since the density of pure copper is around 8900 kg/m³. The visual pigmentation also supports the above fact. Thus, the core material should be copper or an alloy of copper. Furthermore, the silvery appearance, dull grey oxidised surface (iron oxide), and the HRC 37.5 - 60 hardness led to the conclusion of the coated material being consistent with a hard iron/steel plating layer. Thereby the final conclusion regarding the material is as follows;

- Core: Copper alloy
- Outer coating: Iron/Steel
- Tip end: Tin (Sn)

3.2.3 Manufacturing Process

- Core formation: The copper alloy core is produced by casting a copper blank and machining to shape
- Tip shaping: The core is machined into the required tip geometry, conical, in the case of this study.
- Electroplating: A layer of iron (Fe) is electroplated onto the copper alloy core for wear resistance
- Finishing coating: The working end is selectively tinned (Sn coating) for improved solder wetting and usability by electroplating

3.3 Task B – Discussing the feasibility of Casting

- Casting a soldering iron tip is technically possible but presents several significant challenges that make it impractical compared to conventional manufacturing methods.

Justification on the difficulty of employing casting to manufacture Soldering iron tips

Soldering iron tips are industrially made from copper with an iron plating, followed by chrome or tin/solder coatings at the end. Copper is the core material because of its exceptional thermal conductivity. While copper can be cast, the problems arise from:

Shrinkage and porosity - Copper castings are prone to internal voids and gas porosity, which would create hot spots and uneven heat distribution which is fatal for a tip that needs uniform thermal performance.

Dimensional precision - Tips require tight tolerances, especially at the working point. Sand casting and investment casting can't reliably achieve this without extensive post-processing machining anyway.

Surface finish - A cast copper surface is too rough and may become prone to oxidation making it difficult for effective soldering. The functional coatings (iron plating, etc.) adhere much better to a machined/drawn surface.

Scale economics - Tips are small, high-precision parts. Casting setup costs (molds, sprues, risers) don't make economic sense at the scale and geometry involved.

3.4 Task C – Discussing the feasibility of Forming

Soldering iron tips are somewhat feasible to manufacture using forming processes, primarily because copper is highly ductile and formable.

The following forming process might be well suited for the purpose,

- **Cold forging/extrusion**
- **Swaging/rotary forging**

Key advantages of forming can be stated as grain flow alignment, minimal material waste, good surface integrity, and high scalability.

Main limitations are that forming alone isn't sufficient electroplating (iron, nickel, tin layers) and some finish machining are always required afterward to meet functional requirements.

4 Part No 2: Metal Barrel - 230005C

4.1 Introduction

The soldering iron barrel is positioned between the handle and the soldering tip, surrounding the heating element. Its primary functions are to support and secure the soldering tip, protect the internal heating element from damage and oxidation, and assist in heat management through ventilation holes that reduce heat transfer toward the handle. During operation, the barrel is exposed to high temperatures ranging from approximately 300°C to 480°C, repeated thermal cycling, and corrosive conditions caused by soldering flux residues and atmospheric moisture.

4.2 Task A - Product Analysis

4.2.1 Identification of the Material

(i) Pre-inspection

According to straw and blue oxide films formed on the stainless-steel surface, it is clear that the surface was exposed to 290 °C to 590 °C. Magnetic properties exist. Since 300-series stainless steels are generally non-magnetic, a 400-series alloy (which is magnetic) is the primary candidate for a structural heater sleeve.

(ii) Spark Optical Emission Spectrometry Test

Reason for selection

Spark Optical Emission Spectrometry was selected because it provides rapid and highly accurate quantitative elemental analysis for metallic materials, particularly ferrous alloys such as stainless steels.

Results

			1	2	<X>	SD	RSD
C	%	Conc	0.0274	0.037	0.032	<i>0.0067</i>	<i>20.73</i>
Si	%	Conc	0.170	0.218	0.194	<i>0.034</i>	<i>17.45</i>
Mn	%	Conc	0.277	0.288	0.283	<i>0.0082</i>	<i>2.89</i>
P	%	Conc	0.0137	0.0179	0.0158	<i>0.0029</i>	<i>18.57</i>
S	%	Conc	[C<]0.0040	[C<]0.0040	[C<]0.0040	<i>0</i>	<i>0</i>
Cr	%	Conc	12.35	12.32	12.34	<i>0.0194</i>	<i>0.157</i>
Mo	%	Conc	[C<]0.0050	0.0054	0.0052	<i>0.0003</i>	<i>5.33</i>
Ni	%	Conc	0.109	0.113	0.111	<i>0.0034</i>	<i>3.06</i>
Al	%	Conc	[C<]0.0020	[C<]0.0020	[C<]0.0020	<i>0</i>	<i>0</i>
Co	%	Conc	[C<]0.0050	[C<]0.0050	[C<]0.0050	<i>0</i>	<i>0</i>
Cu	%	Conc	0.044	0.043	0.044	<i>0.0008</i>	<i>1.83</i>
Nb	%	Conc	[C<]0.0050	[C<]0.0050	[C<]0.0050	<i>0</i>	<i>0</i>
Ti	%	Conc	0.0115	0.0113	0.0114	<i>0.0001</i>	<i>1.25</i>
V	%	Conc	0.095	0.100	0.097	<i>0.0039</i>	<i>3.96</i>
W	%	Conc	[C<]0.050	[C<]0.050	[C<]0.050	<i>0</i>	<i>0</i>
Pb	%	Conc	[C<]0.0100	[C<]0.0100	[C<]0.0100	<i>0</i>	<i>0</i>
Mg	%	Conc	0.0115	0.0099	0.0107	<i>0.0011</i>	<i>10.65</i>
B	%	Conc	[C<]0.0010	[C<]0.0010	[C<]0.0010	<i>0</i>	<i>0</i>
Fe	%	Conc	86.8	86.8	86.8	<i>0.030</i>	<i>0.035</i>

Conclusion

The results confirmed that the material is a stainless-steel alloy, as the chromium content of 12.34% exceeds the minimum 10.5% chromium requirement necessary for stainless steel classification. The nickel content was found to be very low at 0.111%, indicating that the alloy is non-austenitic and therefore not part of the 300-series stainless steels, which typically contain more than 8% nickel. This conclusion is also supported by the magnetic properties observed in the component.

The measured carbon content was 0.032%, with the remaining composition primarily consisting of iron at approximately 86.8%. These values are consistent with either AISI 410 or AISI 410S stainless steel. However, AISI 410 permits a maximum carbon content of 0.15%, whereas AISI 410S, which is the low-carbon variation developed for improved heat resistance and weldability, has a maximum carbon limit of 0.08%.

Since the measured carbon content is significantly below the 0.08% threshold, the material was identified as AISI 410S stainless steel.

(iii) Salt Test

Method

The specimen was subjected to a salt spray corrosion test using a sodium chloride (NaCl) mist environment to evaluate its corrosion resistance. Prior to testing, one edge of the sample was cut using an angle grinder to expose a fresh metal surface. The specimen was then exposed to the salt spray environment for a specified duration, after which the surface condition and corrosion behavior were visually examined.

Reason for selection

The salt spray test was selected to assess the corrosion resistance of the material under aggressive chloride-rich conditions since the soldering iron barrel operates in environments exposed to atmospheric moisture, heat, and potentially corrosive flux residues, the test was useful in evaluating the stability of the material's protective passive oxide layer and its susceptibility to localized corrosion damage.

Observations



Conclusion

Rusting occurred primarily at the angle-ground cut edge because the grinding process removed the protective passive chromium oxide layer and exposed fresh metal to the chloride-rich environment. The original untreated surfaces showed little to no corrosion, indicating that the intact passive layer continued to provide corrosion resistance.

4.3 Task B - Casting feasibility

Casting is not suitable for the soldering iron barrel because it is a thin-walled perforated component that requires accurate hole formation and uniform thickness. Casting could introduce defects such as porosity and poor dimensional accuracy, while also requiring extensive secondary machining. Precision stamping and roll forming are more suitable as they provide better dimensional consistency, surface finish, thermal performance, and cost efficiency for mass production.

4.4 Task C - Forming

4.4.1 Possibility and Appropriateness

The soldering iron barrel is made from AISI 410S stainless steel and has a thin-walled cylindrical shape with ventilation holes, making it suitable for sheet metal forming processes. Cold working during forming slightly increases strength due to strain hardening, but process control is required to avoid residual stresses that could affect performance under repeated thermal cycling (300°C–480°C).

Bulk forming processes such as forging or extrusion are not suitable due to the thin geometry and perforated structure.

4.4.2 Likely Forming Processes and Selection

Possible processes:

1. Stamping (hole punching)
2. Roll forming (cylindrical shaping)
3. Bending
4. Laser cutting (low volume)
5. Deep drawing (not ideal due to thinning)

Most suitable process:

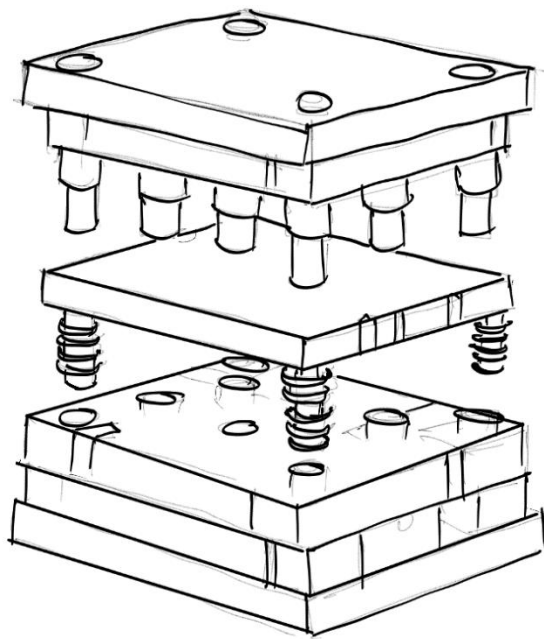
Precision stamping + roll forming

Reasons:

1. Accurate hole production in flat sheet
2. To maintain uniform wall thickness
3. Low cost and high production rate
4. Suitable for mass manufacturing
5. Roll forming avoids thinning and distortion of the sheet metal

Tool Design

For precision stamping



4.5 Identification of the Manufacturing Process

4.5.1 Potential Manufacturing Processes and Justification

Potential Processes:

Manufacturing Process	Suitability
Precision Stamping + Roll Forming	Highly suitable
Deep Drawing	Moderately suitable for seamless cylindrical parts
Tube Extrusion + Secondary Drilling	Suitable but expensive for small components
Spiral Tube Forming	Low suitability due to poor dimensional stability
Laser Cutting + Rolling	Suitable for prototyping and low-volume production

4.5.2 Selected Process – Precision Stamping followed by Roll Forming

Precision stamping followed by roll forming is the most suitable manufacturing method for the soldering iron barrel made from thin AISI 410S stainless steel sheet.

Ventilation holes are first created in the flat sheet using precision stamping with a progressive die. This allows accurate control of hole size, spacing, and repeatability, which is important for consistent airflow and heat dissipation. Producing the holes before forming is also faster and more economical than drilling after the tube is made.

The perforated sheet is then formed into a cylindrical shape using roll forming. This process gradually bends the sheet into a tube without causing significant thinning or distortion, ensuring a uniform wall thickness along the barrel. This is important because the component is exposed to repeated thermal cycling in the range of approximately 300°C to 480°C.

The longitudinal open seam also acts as a controlled expansion gap, allowing thermal expansion and contraction during operation without inducing excessive stresses or cracking.

5 Part No 3: Ceramic Part - 230008M

5.1 Introduction

The ceramic insulation sleeve is a key component in a soldering iron that provides electrical insulation and thermal resistance under high-temperature conditions. It is commonly made from heat-resistant ceramics such as alumina or steatite due to their excellent dielectric strength and thermal stability.

The sleeve is placed around the heating element and internal conductors to prevent electrical leakage, short circuits, and damage during operation. It also provides mechanical support and proper alignment for the internal components. Because of its high heat resistance, mechanical strength, and insulating properties, the ceramic sleeve helps improve the safety, durability, and reliability of the soldering iron.

5.2 Task A - Product Analysis

5.2.1 Identification of the Material

Several standard and non-standard material identification tests were conducted to determine the material sub-class of the ceramic insulation sleeve used in the soldering iron. The tests were selected based on the functional requirements of the component, such as high temperature resistance, electrical insulation, hardness and structural stability. The objective was not to identify the exact material grade but to classify the material under a suitable ceramic sub-category such as alumina based.

(i) Visual Inspection Test (Non-Standard Method)

Method

The component surface, color, texture, and physical appearance were visually examined under normal lighting conditions.

Reason for Selection

Visual inspection is a simple preliminary identification method commonly used in material characterization. It allows rapid identification of characteristic features associated with ceramic materials.



Observations

- White/off-white appearance
- Smooth to slightly rough surface
- Brittle edges
- No visible deformation or corrosion

Conclusion

These characteristics are commonly associated with engineering ceramics used for electrical insulation applications, such as alumina or steatite ceramics.

(ii) Scratch Test (Non-Standard Hardness Evaluation)

Method

The surface of the component was scratched using a hardened steel tip to observe resistance against surface damage.

Reason for Selection

Ceramic materials generally possess high hardness compared to polymers and many metals. This quick comparative method helps distinguish ceramics from softer engineering materials.

Observations

- No visible scratch marks
- High resistance to surface penetration

Conclusion

The high hardness behaviour confirms that the material belongs to the ceramic material subclass.

(iii) Burn / Flame Resistance Test (Non-Standard Thermal Test)

Method

The component was exposed to direct flame / high temperature for a short duration to observe burning, melting, smoke generation, or deformation.



Burn Test

Reason for Selection

The soldering iron component operates in elevated temperatures. Therefore, thermal resistance is a key identifying property. Ceramics typically do not burn or melt under ordinary flame conditions.

Observations

- No burning
- No melting or softening
- No smoke emission

Conclusion

The material exhibits high thermal stability and heat resistance indicating a ceramic insulating material rather than a polymer-based material.

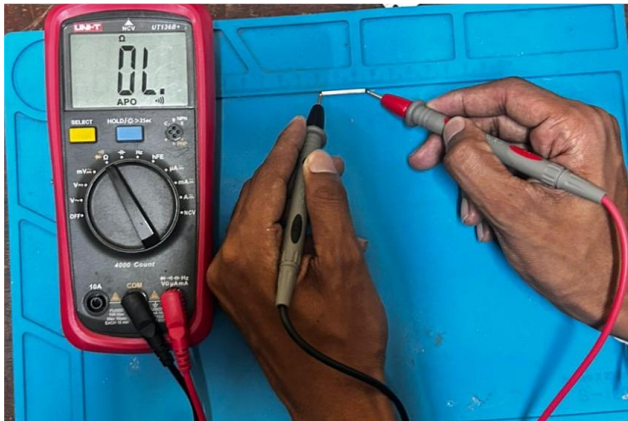
(iv) Electrical Insulation Test (Functional Property Test)

Method

A simple electrical continuity test was carried out using a multimeter to determine whether current could pass through the component.

Reason for Selection

The primary function of the ceramic sleeve is electrical insulation between heating wires. Therefore, electrical insulation behaviour is an essential identifying characteristic.



Electrical Insulation Test

Observations: - No measurable current flow or continuity detected

Conclusion: - The material possesses excellent dielectric properties, confirming that it belongs to the electrical insulating ceramic category.

(v) Density Measurement Test



	Weight (g)
Bottle	16.7685
Bottle + Specimens	17.029
Bottle + Water + Specimen	73.6528
Bottle + Water	73.4609

Density (kg/m ³)	3797.376093
------------------------------	-------------

Density Test

Reason for Selection

The water displacement method was selected because the ceramic sleeve has a hollow cylindrical geometry, making direct volume calculation difficult and less accurate. This method provides a simple and reliable approach for estimating the density of irregular ceramic components.

Observations

- Measured density $\approx 3797 \text{ kg/m}^3$
- No water absorption or visible reaction during testing

Conclusion

The measured density is close to the typical density range of alumina ceramics, indicating that the material likely belongs to the alumina-based engineering ceramic sub-class commonly used for electrical insulation and engineering ceramic sub-class commonly used for electrical insulation and high-temperature applications.

Property	Unit	AL ₂ O ₃ 99.7	AL ₂ O ₃ 99.5	AL ₂ O ₃ 99	AL ₂ O ₃ 95
Purity	—	99.7%	99.5%	99%	95%
Density	g/cm ³	3.92	3.9	3.8	3.7
Bending Strength	MPa	375	370	340	304
Compressive Strength	MPa	2450	2300	2250	1910
Elastic Modulus	GPa	380	370	330	330
Fracture Toughness	MPam ^{1/2}	4.5	4.3	4.2	3.8
Hardness	HRA	91	91	90	89
Vickers Hardness	HV1	1600	1550	1450	1400
Thermal Expansion Coefficient	10 ⁻⁶ K ⁻¹	7.8	7.8	7.7	7.5
Thermal Conductivity	W/mk	32	32	25	25
Thermal Shock Stability	ΔT, °C	220	220	200	200
Maximum Operating Temperature	°C	1750	1750	1700	1500
Volume resistance at 20°C	Ω·cm	10 ¹⁵	10 ¹⁵	10 ¹⁴	10 ¹⁴

5.2.2 Final Conclusion, Material Identification and Justification

The measured density value of approximately 3797 kg/m³, together with its high hardness, excellent electrical insulation, and resistance to high temperatures, closely matches the properties of high-purity alumina ceramics used in soldering iron applications. Therefore, the material is concluded to be a high-purity alumina ceramic, likely within the range of 95%–99% alumina (Al₂O₃) purity, commonly used for electrical insulation and thermal protection in high temperature electronic devices.

5.3 Task D - Non-Metallic Manufacturing

5.3.1 Possibility and Appropriateness

The ceramic insulation sleeve used in the soldering iron is manufactured from high-purity alumina ceramic due to its excellent electrical insulation, thermal resistance and mechanical stability at elevated temperatures. Since ceramics are brittle materials with very low plastic deformation capability, conventional metal forming processes are not suitable for manufacturing this component. Similarly, polymer and rubber processing methods are inappropriate because the component operates under high-temperature conditions that exceed the thermal limits of those materials.

Therefore, ceramic powder forming processes followed by sintering are required for manufacturing the component. Among ceramic forming methods, dry pressing is considered the most appropriate process for this application due to the small size, simple geometry, and dimensional accuracy requirements of the ceramic sleeve.

Suitable Manufacturing Method – Dry Pressing

In the dry pressing process:

- Fine alumina ceramic powder mixed with a small amount of binder is placed into a steel die.
- High pressure is applied using a punch to compact the powder into the required shape.
- A central core pin is used to form the internal hole of the sleeve.
- The compacted “green body” is removed from the die.
- The component is then sintered at high temperature to achieve final strength and density.

Changes to Material Properties During Manufacturing

- Powder Compaction - Increases particle packing density
- Green Body Formation - Provides temporary handling strength
- Sintering - Increases hardness, density, and strength
- Final Ceramic State - Develops high thermal resistance and electrical insulation

During sintering

- Porosity decreases
- Density increases
- Mechanical strength improves
- Electrical insulation properties improve

However, the material also becomes more brittle after sintering.

5.3.2 Potential Manufacturing Processes and Justification

Manufacturing Process	Suitability
Dry Pressing	Highly suitable
Ceramic Extrusion	Suitable for long continuous tubes (Not Hollow)
Slip Casting	Suitable for complex hollow shapes
Ceramic Injection Molding	Suitable for complex small components

Selected Process – Dry Pressing

Dry pressing is proposed as the most suitable manufacturing method for the ceramic insulation sleeve.

Justification

- Suitable for small ceramic components with simple geometry
- Provides good dimensional accuracy
- Economical for mass production
- Produces high-density ceramic components
- Commonly used for electrical insulating ceramics
- Internal hole can be formed using a core pin during pressing
- Requires minimal post-processing after sintering

Therefore, dry pressing followed by high-temperature sintering is considered the most appropriate manufacturing process for the ceramic insulation sleeve used in the soldering iron.

6 Part No 3: Plastic Handle - 230393K

6.1 Component Overview

The handle of a soldering iron is a critical safety and ergonomic component. It serves as the primary interface between the user and the tool. Its functions are to protect the user from electrical shock, provide thermal insulation to prevent heat transfer to the hand, and house the internal wiring connections. Furthermore, it must offer a comfortable grip for safe, precise, and fatigue free handling during continuous operation. This section details the reverse engineering of the handle, including material identification testing, manufacturing process identification and proposed design improvements.



6.2 Task A - Product Analysis

6.2.1 Identification of the material

To identify the material class of the soldering iron handle, a series of standard and non-standard physical tests were conducted.

(i) Visual Inspection (Non-standard Test)

A visual and mechanical inspection revealed that the handle possesses a shiny, scratch-resistant surface finish.

Reason for selection

To observe the surface quality, appearance, and basic mechanical behavior of the material before conducting detailed tests.

Conclusion

Under a bending load, the material exhibited brittle fracture (shattering).

(ii) Density Test

A water displacement test (Density Bottle) was conducted.

Reason for selection

This was chosen because density is a key property for material classification. Engineering plastics generally have much lower density than metals and ceramics. The sinking behavior and calculated density help confirm whether the material falls within the typical polymer range.

	Weight (g)		
Bottle (W)	16.7685	Density (kg/m ³)	1338.116032
Bottle + Specimens (W1)	16.983		
Bottle + Water + Component (W2)	73.6351		
Bottle + Water (W3)	73.5809		



Conclusion

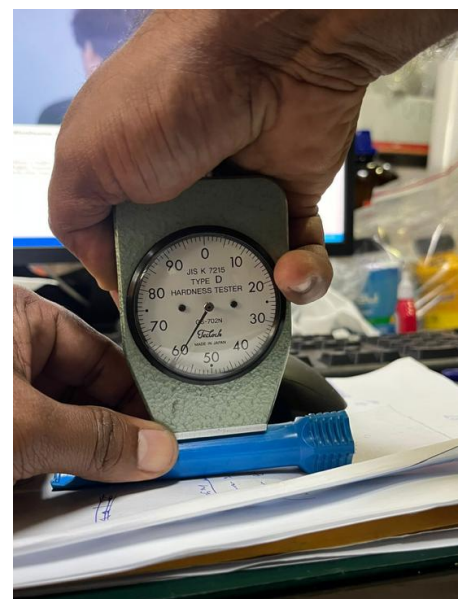
The sample sank, and the calculated density was recorded at 1338.12 kg/m³.

(iii) Shore D Hardness Test

The surface hardness was measured at Shore D = 60.

Reason for selection

This was used to quantitatively measure surface hardness. Shore D is specifically suited for harder plastics and helps distinguish engineering thermoplastics used in structural applications from softer polymers. It provides a numerical value for comparison with standard material ranges.



(iv) Burn Test

When exposed to a flame, the material exhibited clear thermoplastic behaviour. It softened initially and melted gradually without significant, rapid dripping. As it burned, there was a noticeable colour change, and the tubular structure deformed and collapsed.

Reason for selection

This was selected to observe thermal response and combustion behavior. Thermoplastics soften and melt gradually when heated, unlike thermosets or metals. Observing softening, deformation, and melting behavior helps confirm that the material is a thermoplastic polymer.



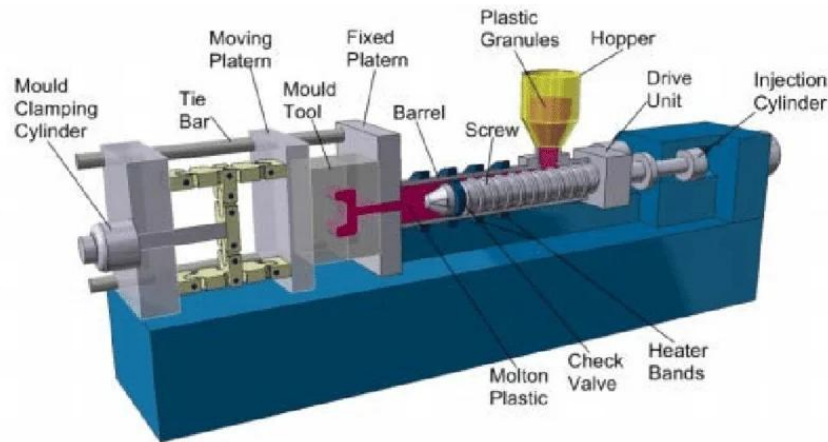
6.2.2 Final Conclusion, Material Identification and Justification

The pre-inspection results (scratch resistance and specific fracture behavior), combined with the density and hardness values, fall well within the typical range for engineering plastics.

The burn test confirmed its thermoplastic nature. Based on these composite results, the material subclass is identified as **ABS (Acrylonitrile Butadiene Styrene)** with some additives included to improve its functionalities.

6.2.3 Manufacturing Process

The most likely manufacturing process utilized for the current handle is **Injection Moulding**.



Manufacturing plastic handle using Injection Moulding Process:

- ABS plastic pellets are fed into the hopper.
- The pellets move through the heater bands by a rotating screw mechanism and melt due to heat.
- The mould closes tightly using a clamping unit.
- Inside the mould, core pins and core inserts are positioned to create internal features.
- The rotating screw injects molten metal into the mould cavity.
- The molten plastic flows around the core pins and inserts, forming both external and internal shapes of the handle.
- Holding pressure is applied to completely fill the cavity.
- After cooling the mould opens and the core pins separate from the component, leaving the internal holes and cavities.
- Ejector pins push the handle out of the mould.

Justification

The soldering iron handle features a complex 3D geometry with smooth, ergonomic curves that would be difficult and costly to machine. Injection moulding is highly suited for processing ABS, as the thermoplastic melts and flows smoothly into intricate mould cavities. This process naturally yields the excellent, shiny surface finish observed during the pre-inspection. Furthermore, injection moulding is the industry standard for producing high-

durability polymer housings, offering high dimensional accuracy, repeatability, and cost-effectiveness for mass production with minimal post-processing.

6.3 Task D - Non-Metallic Manufacturing

6.3.1 Possibility and Appropriateness

Possibility

Manufacturing the handle using polymer shaping processes is highly feasible. The industry has a robust supply chain for ABS pellets, and modern tooling can easily accommodate the internal hollowing required for wiring and structural integrity.

Appropriateness

Utilizing a polymer shaping process like injection moulding is perfectly appropriate for this application. It effectively delivers the required material properties, specifically high electrical resistance and thermal insulation that are mandatory for a soldering iron handle.

Changes to Material Properties

During thermal shaping processes, the polymer chains align along the direction of the material flow, which can result in slight anisotropic strength properties. Additionally, rapid or uneven cooling inside the mould can induce residual thermal stresses, which must be carefully managed through optimized cooling channels in the mould design.

6.4 Potential Manufacturing Processes and Justification

Potential Processes

1. Injection Moulding
2. Blow Moulding
3. Polymer Extrusion
4. 3D Printing (Fused Deposition Modelling)

Most Suitable Process: Injection Moulding.

Justification

While 3D printing is excellent for initial prototyping and extrusion is suitable for continuous, uniform cross-sections (like pipes), injection moulding is the only practical and economical method for mass-producing the variable wall thicknesses, internal mounting features, and closed-end 3D geometry of the handle. It provides the tightest tolerances and best aesthetic finish.

6.5 Proposed Modification and Improvements

While standard injection moulding is efficient, the current single-material design can be optimized. We propose utilizing an Overholding process. By injecting a softer rubber material, such as TPE (Thermoplastic Elastomer), over the rigid ABS core, the handle will gain a superior, slip-resistant grip. This modification, alongside an optimised surface texture and an improved ergonomic shape, will significantly reduce hand fatigue and improve overall control during precision soldering tasks.

7 Part No 5: Heating Element - 230120A

The component analysed in this section is the Heating Element of a standard 230V soldering iron. It is the primary energy conversion component of the assembly, responsible for converting electrical energy into heat via resistive (Joule) heating to raise the metal tip to soldering temperature ($\sim 450^{\circ}\text{C}$).

The element consists of a fine-gauge resistance wire wound in a tight helical coil around a ceramic tube former, with thicker copper terminal leads welded at each end. It operates continuously at mains voltage whenever the iron is plugged in, making material selection and manufacturing precision critical to both performance and safety.

The core functions of the heating element are:

- Primary Energy Conversion
- High-Temperature Generation
- Thermal Stability

2.1 Identification of the Material

(i) Visual Inspection Test

Visual inspection of the heating element revealed a fine-gauge wire ($\sim 0.35\text{mm}$ diameter) wound in a tight helical coil around a ceramic tube. The wire exhibited a dull grey-silver colour with no metallic sheen, and slight surface oxidation consistent with repeated high-temperature exposure. These observations initially pointed toward Nichrome 80/20 (80% Nickel, 20% Chromium) as the base material, ruling out copper (reddish colour) and stainless steel (brighter finish).



(ii) Density Test

A water displacement test (Density Bottle) was conducted. A sample of wires were used and it sank. The density was calculated according to the following observations.

Results

- Bottle weight: 16.7611g
- Bottle + Specimen: 16.8632g
- Bottle + Water + Component: 73.6609g
- Bottle + Water: 73.5717g
- **Measured Density: 7914.73 kg/m³**

This was cross-referenced against known material densities:

- Copper: 8960 kg/m³
- Aluminium: 2700 kg/m³
- Kanthal (FeCrAl): ~7100 kg/m³ ; (obtained value was ~12% higher).
- Nichrome 80/20: 8200–8400 kg/m³



(iii) V-I Characteristic Test

A DC power supply was connected across a 15cm element sample of wire. Voltage was stepped to 12V and 24V, with current measured via ammeter and wattage read from the supply display. Resistance was back calculated using $R = V/I$.

Voltage (V)	Current (A)	Power (W)	Calculated R (Ω)
12	0.247	2.96	48.58
24	0.505	12.12	47.52

Current scaled linearly with voltage with no erratic fluctuations, confirming pure Ohmic behaviour. This rules out semiconducting or composite wire materials. The slight resistance increases from cold to hot is consistent with a positive but low TCR which is a characteristic of Nichrome.



(iv) Cold Resistance & Temperature Coefficient of Resistance Estimation

Cold resistance was measured directly using a multimeter at room temperature (30°C) before any power was applied.

- Cold Resistance: 46.6Ω at 30°C
- Voltage of 12V: 48.58Ω at 102°C
- Voltage of 24V: 47.52Ω at 238°C

TCR calculated as $\alpha \approx 9.18 \times 10^{-5} / ^\circ\text{C}$

TCR value for Nichrome 80/20 is approximately $\alpha \approx 11.8 \times 10^{-5} / ^\circ\text{C}$



The experimental result is within the same order of magnitude as the reference value. The difference is likely due to open-air testing conditions and heat losses at the thermocouple contact points, which can affect temperature measurements. Overall, the results are consistent with the expected behaviour of Nichrome heating elements.

Material Conclusion - Process of Elimination

The measured density of 7914.73 kg/m³ ruled out common metals such as copper and aluminium, while standard iron was eliminated due to its poor high-temperature oxidation resistance. Kanthal was also considered unlikely because its density is lower than the measured value.

The V-I test confirmed stable Ohmic behaviour, and the calculated TCR value of 9.18×10^{-5} /°C was consistent with the behaviour of Nichrome alloys. The dull grey surface and stable oxide layer also matched the typical characteristics of Nichrome.

Based on the density, electrical properties, and oxidation behaviour, the material was identified as Nichrome 80/20.

7.1 Identification of the Manufacturing Process

Observations from Disassembly

- Wire diameter was uniform throughout the entire coil
- Coil spacing was consistent with no visible irregularities
- No weld marks were seen along the coil itself, only near the terminal connections
- Terminal leads were thicker than the heating wire and joined at the ends of the coil
- Ceramic tube surface was smooth and showed no machining marks, suggesting a sintered manufacturing process

Possible Manufacturing Methods

1. Wire drawing used to produce the thin Nichrome wire from rod stock
2. Mandrel coil winding used to form the helical shape around the ceramic tube
3. Resistance welding used to join the terminal leads to the heating coil
4. Annealing carried out after drawing to reduce internal stresses and restore ductility before winding

Selected Method: Wire Drawing followed by Precision Mandrel Coil Winding

Nichrome rod stock is pulled through a series of progressively smaller tungsten carbide dies to reach the target diameter of 0.35mm. The drawn wire is then wound around a rotating ceramic tube using a CNC-controlled pitch winding machine, ensuring turns do not touch (preventing short circuits) while maximising heat density per unit length. Thicker copper terminal leads are then fused to the coil ends using high current resistance welding as standard solder cannot be used with Nichrome.

Annealing is performed after wire drawing to relieve work-hardening stresses introduced during the drawing process, restoring sufficient ductility for winding without cracking.

7.2 Task B - Casting feasibility

Casting is not a suitable manufacturing method for the Nichrome heating element wire. The required geometry, dimensional tolerance, and microstructural requirements all necessitate a forming-based approach (wire drawing) rather than casting.

7.3 Task C - Forming feasibility

Wire drawing is the main forming process used to manufacture Nichrome heating element wire, and it is well suited for this application. In this process, Nichrome rod stock is pulled through a series of progressively smaller tungsten carbide dies to reduce the diameter and increase the length of the wire. Several drawing passes are needed to reach the final diameter of about 0.35 mm. Since cold working causes strain hardening and reduces ductility, intermediate annealing stages are carried out between passes to prevent cracking.

Wire drawing is suitable because it produces a very uniform wire diameter, which is important since the electrical resistance depends on the wire's cross sectional area. Any variation in diameter can cause uneven heating and local hotspots. The process also provides a smooth surface finish, helping reduce oxidation and improving the service life of the heating element.

During drawing, the wire becomes stronger and harder due to work hardening, but less ductile. A final annealing process at around 800–1000°C is therefore performed before coil winding to relieve internal stresses and restore ductility without significantly affecting the electrical resistivity of the Nichrome alloy.

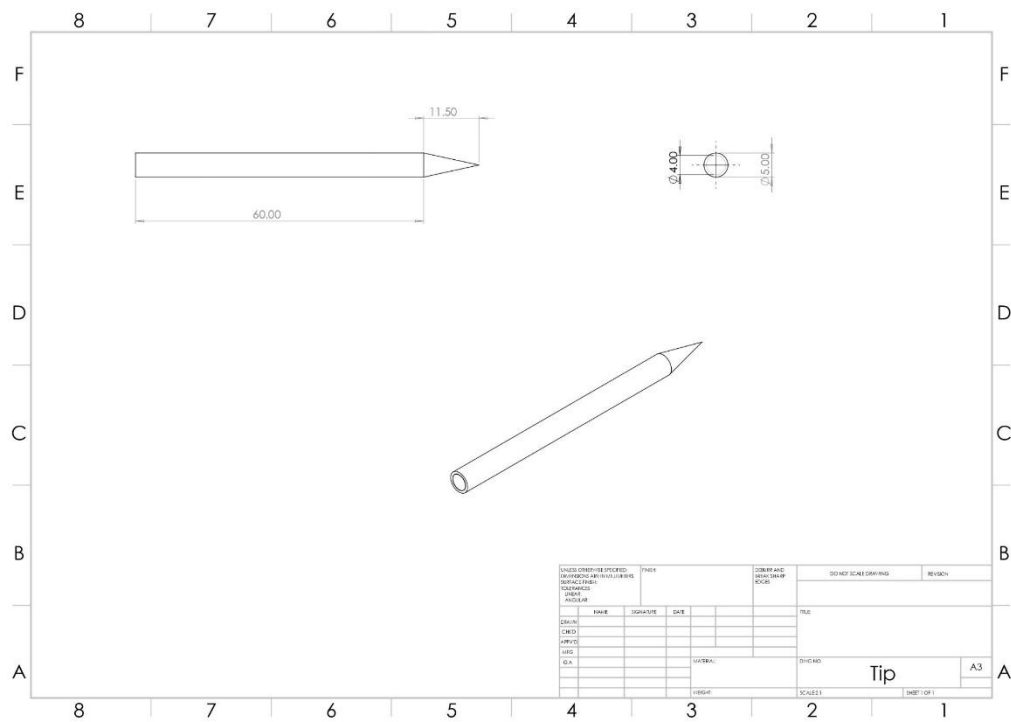
After drawing and annealing, the wire is formed into a helical coil by mandrel winding around a ceramic tube under controlled tension. The coil pitch must be carefully controlled so that adjacent turns do not touch, as this could create short circuits and overheating. Together, wire drawing and coil winding form the complete manufacturing process for the heating element.

7.4 Non-Metallic Manufacturing

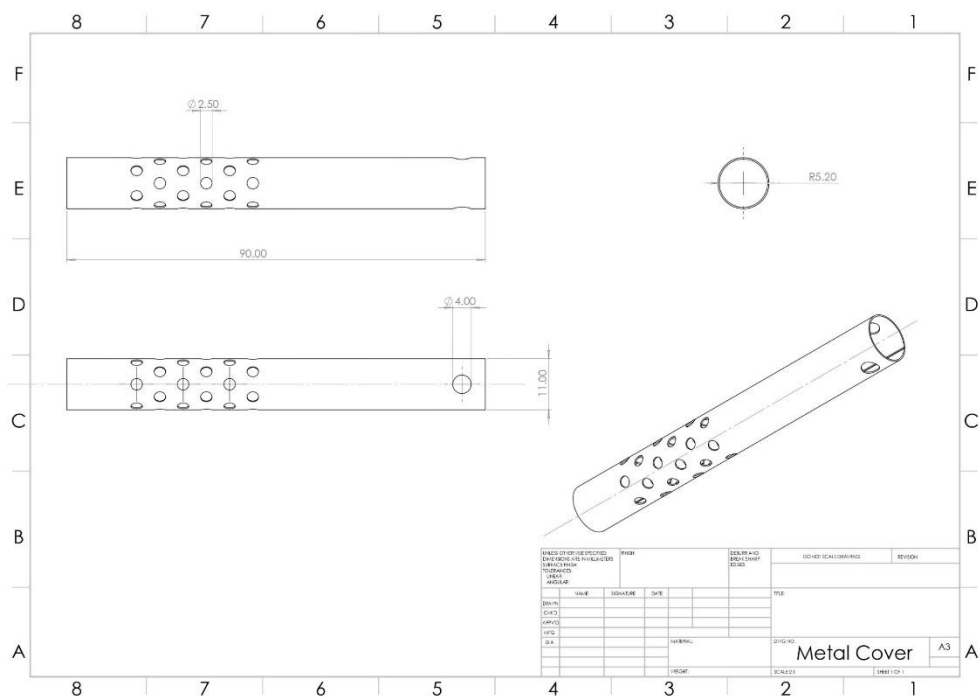
Non-metallic manufacturing processes are not suitable for the Nichrome heating element wire because Nichrome is a high-temperature metal alloy.

8 Production Drawings

1. Soldering Iron Tip



2. Metal Barrel



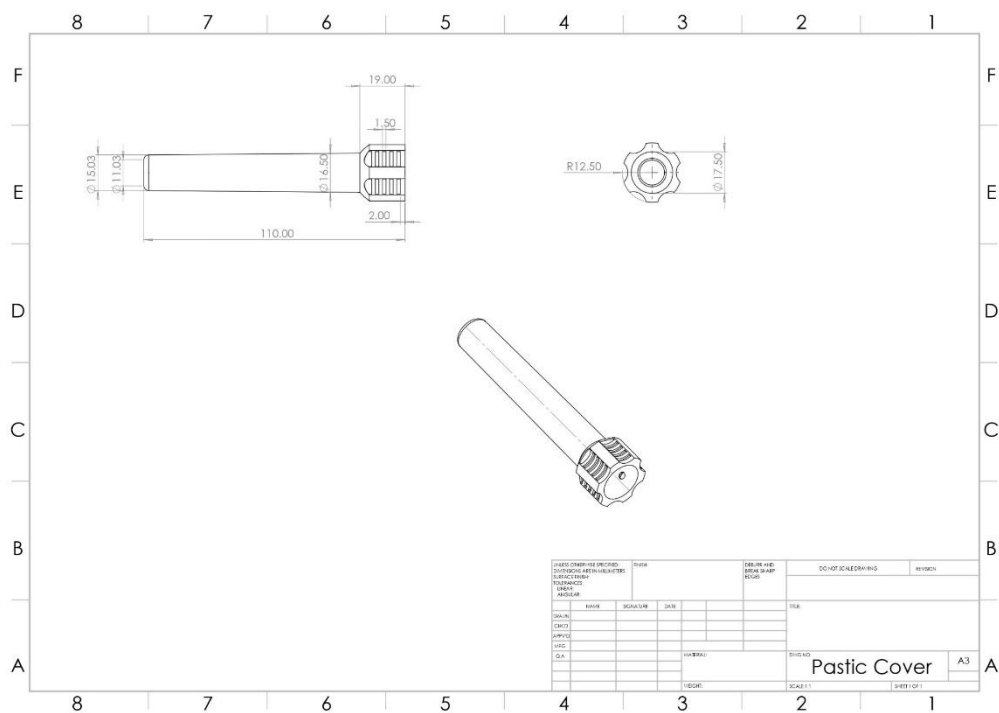
3. Heating Element

Technical drawing of a heating element showing three views: front, top, and isometric. The front view is a rectangle with a width of 10.00. The top view shows a circular cross-section with a diameter of 0.35. The isometric view shows the element as a cylinder. The drawing includes a title block with the title 'Heating Element' and a sheet number 'A4'.

4. Ceramic Component

[illegible]

5. Plastic Handle

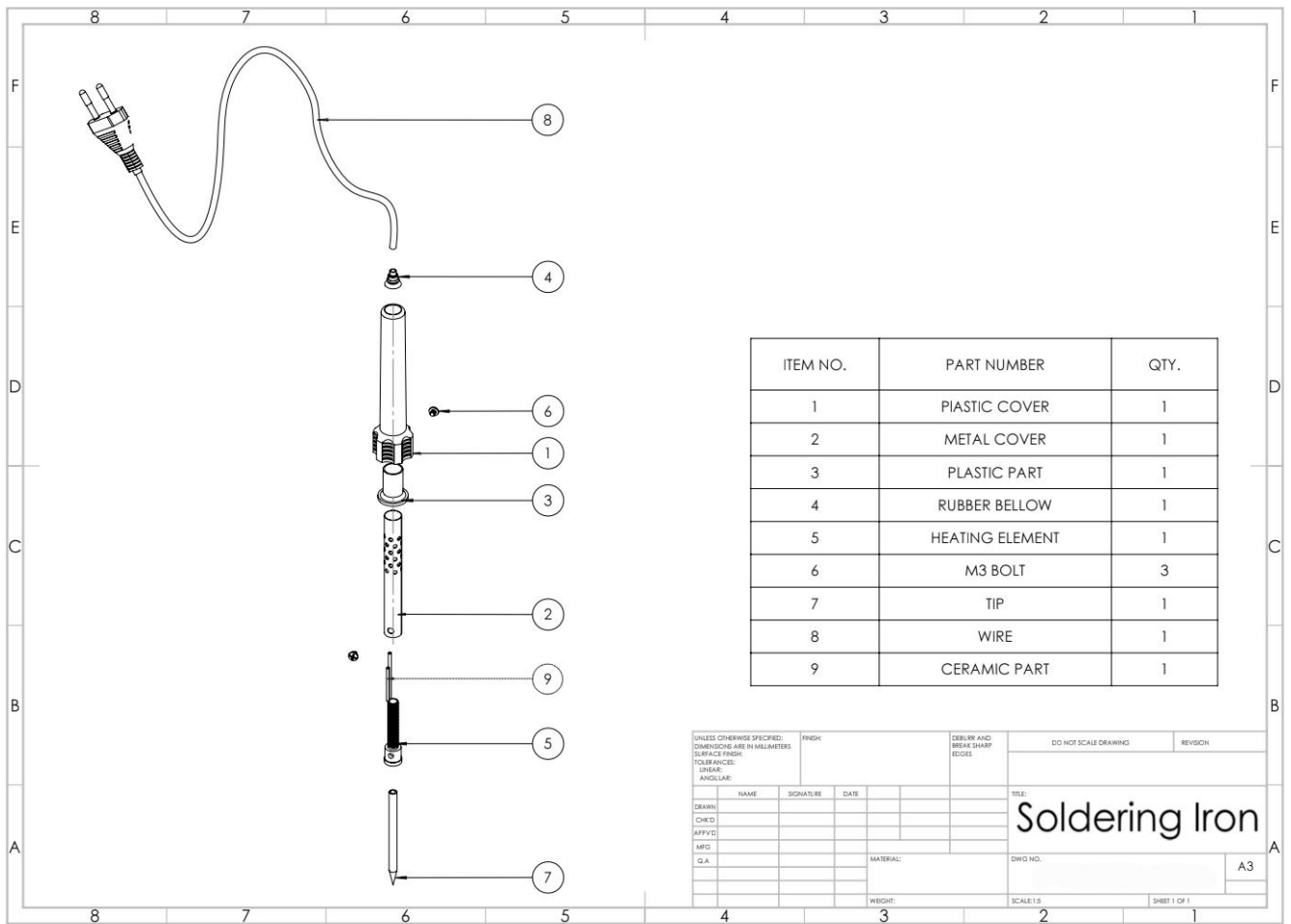


Product Assembly / Joining Soldering Iron

Assembly Order	Component	Assembly/Joining Process	Justification
1	Heating Element with Ceramic Insulation Sleeve	Insertion assembly	Provides accurate positioning and insulation without damaging the brittle ceramic part.
2	Mica Sheet	Wrapping	Easily covers the heating element and provides effective thermal and electrical insulation.
3	Metal Cover (Barrel)	Mechanical fastening	Gives strong protection and allows easy assembly and maintenance.
4	Soldering Tip	Press fit or screw fastening	Ensures firm contact for efficient heat transfer and easy replacement.
5	Plastic Handle	Screw fastening	Provides secure attachment while allowing safe handling and easy disassembly.
6	Electrical Wire Connections	Soldering process	Creates reliable electrical conductivity and strong electrical connections.

The selected assembly processes are suitable because they provide proper alignment, electrical insulation, mechanical strength, heat transfer efficiency and ease of maintenance while preventing damage to sensitive components such as the ceramic insulation sleeve.

Exploded view of Assembly



9 References

“Metallic heating elements,” [Kanthal](https://www.kanthal.com/en/products/furnace-products/electric-heating-elements/metallic-heating-elements/). [Online]. Available: <https://www.kanthal.com/en/products/furnace-products/electric-heating-elements/metallic-heating-elements/>. [Accessed: May 8, 2026].

“Everything you should know about nichrome wire,” *EHISEN*, Mar. 28, 2024. [Online]. Available: [EHISEN Nichrome Wire Article](#). [Accessed: May 11, 2026].

“Nichrome Wire Tech Tips,” *WireTronic Inc.*, Jul. 17, 2020. [Online]. Available: [WireTronic Nichrome Wire Tech Tips](#). [Accessed: May 12, 2026].

Alumina Sleeve,” GORGEOUS CERAMICS. [Online]. Available: <https://ggsceramic.com/product/alumina-sleeve.html>

“Alumina Ceramic Sleeve,” Duratec Ceramic. [Online]. Available: <https://www.duratecceramic.com/alumina-ceramic/alumina-ceramic-tube/alumina-ceramic-sleeve.html>

“Alumina Shaft Sleeve (Al₂O₃ Ceramic Shaft Sleeve),” CSCERAMIC. [Online]. Available: https://www.csceramic.com/alumina-shaft-sleeve-al2o3-ceramic-shaft-sleeve_p1103.html

<https://okboxer.com/abs-vs-other-plastic-handles-which-should-you-choose/>

OK Boxer. (n.d.). ABS vs Other Plastic Handles: Which Should You Choose?* Retrieved from <https://okboxer.com/abs-vs-other-plastic-handles-which-should-you-choose/>